Leopard - Stakeholder Interview

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Version 1
COMPSCI 4ZP6
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This report documents details and outcomes of our meeting with Dr. Denise Geiskkovitch, an assistant professor at the department of Computing and Software at McMaster University. Denise has taught COMPSCI 1MD3: Introduction to Programming for multiple semesters, making her an optimal stakeholder to provide feedback for our platform. She has valuable insight on how programming courses are taught, and what an ideal application would be. Dr. Geiskkovitch is also an expert in human-computer interfaces, so she is able to provide us great insight on the usability of our user interface as well.

This stakeholder interview involved an introduction to the project, a walkthrough of both the student side and teacher side, and letting the interviewee access and play around with the platform. This took approximately 90 minutes.

We summarize the results of this interview as tasks we took away to improve our platform, followed by an appendix of the raw-notes taken by group members during the interview.

Takeaways and To-dos

Miscellaneous

- [low, wont fix] Show student work activity over time:
    - timestamps of when the student worked
    - timeline/scrubber
    - inactivity or time-skip flags
    - aggregate activity graph with density/interest peaks
- [part of other problems] Separate **Save** and **Publish** actions where relevant.
- **[discussed in UI design]** Improve accessibility and clarity:
    - larger text
    - bolder button text
    - more obvious/clickable buttons
    - clickable elements should look interactive
    - reduce excess padding/margins
    - prefer blue over pink
    - support zoom levels
- **[Will fix]** Add better table controls:
    - sorting
    - filtering
    - Searching
- **[Krish] Default Sorting Arrow**

- **[High Priority, low complexity] Have a way to change role to teacher**
    - https://app.graphite.com/github/pr/Beanie-Brick-Band/leopard/316/feat-admin-role-page Krish has open PR related

Create Assignment Form

- **[Low priority, low complexity, Rachelle]** Reorder Fields (professor preference):
    - Name
    - Due Date
    - Description
    - Release Date
    - Consider placing the date / release date below the assignment name in the UI to match Avenue (professors are used to avenue).
- **[Low priority, high complexity, Tim]** Allow assignment grading scheme configuration:
    - weight
    - points / “out of X”
- **[High priority, low complexity, Felix]** Fix release-date behavior: (high priority)
    - release date is currently broken
    - assignments should not be viewable before release date
    - future release times should work correctly
    - Likely no implementation yet

Professor Assignment Management Page

- **[high priority, low/mid complexity, Krish]** Add student assignment search/sort table:
    - by name
    - submission time
    - student ID
    - Email
- **[High priority, mid complexity, Arian]** Add **Download Submissions** button:
    - source code
    - individual and batch download
- **[wontfix]** Download/share Replay
    - replay video format where applicable
- **[mid priority, mid complexity, Krish]** Add assignment description enhancements:
    - syntax highlighting
    - image/file/PDF import support
    - Add a pdf viewer to assignment page + pdf upload for teachers

    - Add label to starter code upload UI component that you can upload PDF as well
- **[Won't fix, address by download button]** Add instructor tooling:
    - This needs design consideration not trivial
    - run student code from a snapshot
    - terminal access for individual submissions
    - terminal/script runner for all student assignments
    - upload and run autograding scripts
        - Submission grading
        - Rigid autograding where starter code is in specific format and
- **[high priority, mid complexity, Krish]** Add suspicious submissions review area for instructors.
    - Add suspicious-work flagging for submissions, visible only to instructors/TAs.
    - flagging assignment as a whole, not specific timestamps in replay
    - new table column
    - flag button
- **[mid priority, low complexity, Felix]** Remove pending enrollment UI.
- **[wontfix]** Add McMaster login / McMaster email-based signup.

Student View

- **[mid priority, low complexity, Tim]** Show assignment due date with time.
- **[mid priority, low complexity, Tim]** Show assignment submission time.
- **[mid priority, low complexity, Tim]** Show assignment last edited time so students can understand progress over time.
- **[high priority, low complexity, Felix]** Allow students to see their grade.
- **[high priority, low/mid complexity, Rachelle/Arian (needs downloadability for prt 1)]** Add preview of code/solution before submission.
    - Can be a snapshot of replay at the final stage. Change replay to have that option
    - Step: 1 get a downloadable final submission similar to ta/prof view
    - Step 2: inline view of code without need for download
- **[high priority, low complexity, Arian]** When submitting an assignment close down Coder workspace to prevent further changes. Prevent launching workspace again
- **[mid priority, low complexity, Arian]** Open a new BROWSER tab when opening the codespace

Review Submission

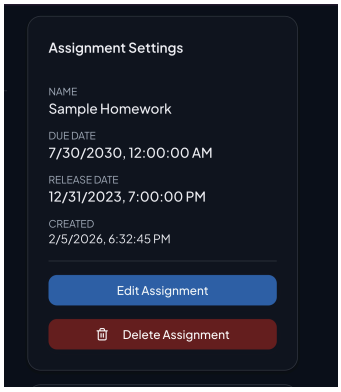
- **[mid priority, mid complexity, Rachelle]** Add next/previous submission navigation.
- **[high, easy, Tim]** Rename **Current Grade** to **Grade**.
- **[high, easy, Tim]** Prevent invalid grading input:
    - no negative grades
    - remove arrow steppers on grade input
- **[high priority, mid, Felix]** Implement two separate actions for grades: **Save** and **Publish (make visible) Grades**.“ Publish” should be a distinct action, not just implied by save/update.
- **[wont fix]** Show instructor-facing activity insights during review:
    - flag
    - skipping / time gaps
    - timestamps
    - density/activity graph
    - suspicious behavior markers

Assignment Creation / Release / Publishing

- **[fixed by release date]** Release date and publishing flows should be clearer and separate where needed.
- **[high, easy, Felix]** Release and due-date UI should better communicate timing. **(only shows date and not time right now)**

Frontend / UI [mid priority, mid complexity]

- **[mid priority, mid complexity, Rachelle]** Improve navbar styling. (colours on navbar dont make it appear clickable as well as location) MAKE BREADCRUMB, blue or on hover blue somehow make it blue bro
- **[mid priority, mid complexity, Krish]** Make buttons feel more clickable. (colour, sizing) (ex. Created text smaller than release date and button text)



- **[mid priority, mid complexity, Krish]** Increase overall text size.
- **[mid priority, mid complexity, Felix]** Reduce unnecessary spacing between assignment elements.
- **[mid priority, low complexity, Krish]** make tables bonita (on dark mode mainly)

Open / Rapid-Fire Ideas

- Add a preview of the students code in submission screen
    - Can be a snapshot of replay at final stage
- **[wont fix]** Add auto-grading visibility in both student list and individual student views.

Appendix

Krish's Notes

- Add flagging button inside of assignment for suspicious work so professors/teaching assistant can tag parts of replays
- Show times of when student worked on it
- activity aggregate graph (like youtube interest etc)
- Publish grades instead of just show
- syntax highlighting for assignment description
- download submissions button
- run code button? Or like spin up an instance for the TA/Teacher
- mcmaster login button(?)
- zoom levels
- table filter / sorting
- remove pending enrollments UI
- departure/arrival UI for release date/due date
- RELEASE DATE IS BROKEN
- save and update are different actions ("Publish" action)
- Rapid fire

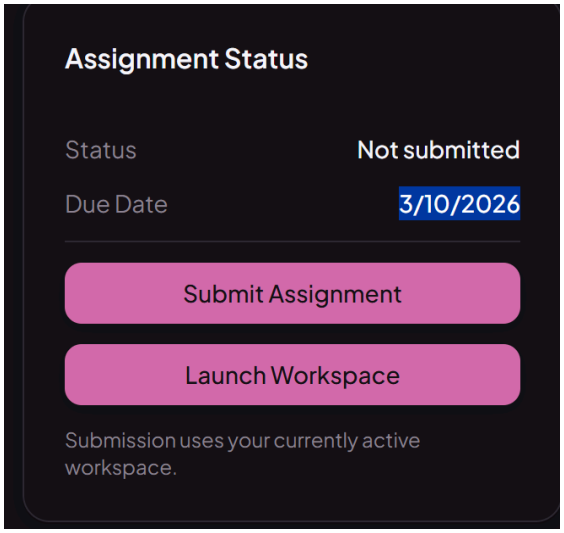
Create Assignment Form:

- Name
- Due Date
- Release Date(?)

Timothy Notes:

- download submissions button (video format + code)
- navbar make it look like an actual navbar

- student sorting and searching
- marks / grading scheme (weight)
- due date should show time for students



Rachelle's Notes

Misc Points
- Add mac id to the student submission by adding mac id signup (?)

Prof Assignment Management Page
- Way to search student assignments and sort (name, submission time) just table it
- file/image import to assignment description
- Main assignment page has prof terminal to do commands on all student assignments
- make section to review all suspicious flagged submissions

Student View
- Student Assignment view should have submission time
- Allow student to see their grade
- add preview of code for student before they submit

assignment creation improvements
- put due date below name (if we want)
- potentially add way to have "out of X grades"
- release time being later than present doesn't work

Review Submission
- remove "status", change current grade to grade,
- Save and publish grades should be separate
- inform instructor of time (flags, skipping, showing the time, density, interest peak)
- grade can be input negative, remove arrows
- Way to run student code from snapshot, and use terminal

- make way to flag button within assignment submission page for instructor to flag smth as suspicious for further review
- Add next/previous submission from review

front end
- bold button text
- text could be generally larger
- reduce padding/margins between assignment
- buttons should have way to appear more clickable
- clickable things should not look like non clickable things
- BLUE better than pink

Felix Notes

1. make a preview of the solution
2. tagging assignments for suspicious behavior. only visible to teacher
3. make new tab of editor the default
4. display to editor information of when it was last edited to let student understand their progression
5. scrubber to show time
6. timeline flags to show when there's timeskips. show activity through the density of a graph. have interest peaks on youtube
7. show data on when students are working on their assignments
8. remove arrow buttons for grade
9. save and publish as separate steps for assignment submissions
10. download button for source code and replay
11. auto grade on the student list side and individual student side. have a terminal that can have a script you can upload and run on every student
12. search by student, student id, email
13. sign up with mcmaster email
14. can make text larger for accessibility
15. increase text size
16. make navbar more obvious
17. remove padding and margins
18. put release and due date below assignment name
19. grade in terms of point scheme
20. don't let assignment be viewed before release date
21. assignment editor should have images or pdf
22. syntax highlighting for editor
23. due date time in assignment view date
24. submission details: current grade → grade. show status (published, submitted, etc.) as well. don't need status view
25. make publish (teacher and release be at separate times
26. make the review submission view a modal carousel